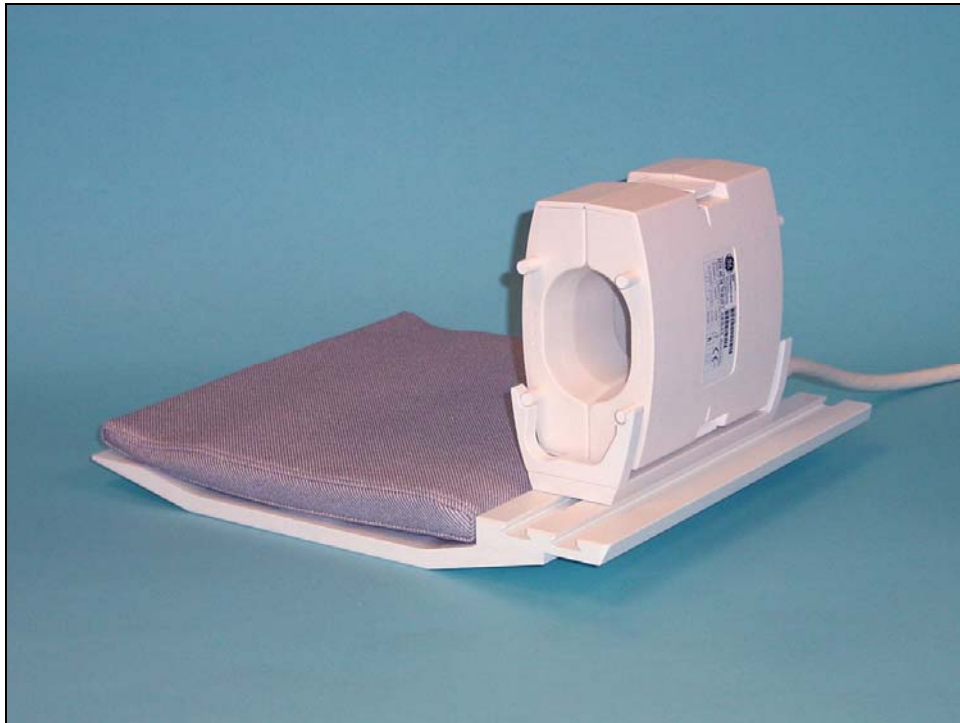


Operator's Manual

1.5T HD 4 Channel Wrist Array for the GE Value MR System



Manufacturer
Invivo
Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608 U.S.A.
500190 Rev 05

CE 0123

Distributed by:
GE Healthcare



GE Part Number 5145565-3

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Licenses and Trademarks

The Invivo Logo is a registered trademark of Invivo Corporation. "Value" is a registered trademark of the General Electric Company.

Proper performance of this coil is guaranteed only while the coil is being used on the MR system (hardware/software level) specified at the time of purchase. Upgrades or other modifications to the system software and/or hardware may affect compatibility. Prior to upgrading your MR system, please contact your GE Healthcare representative to discuss coil compatibility issues. Failure to do so may void your warranty.

Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:

European Representative

GBM Authorised Representative Ltd.
The White House, 2 Meadow
Godalming, Surrey GU7 3HN
United Kingdom



 Attention, Consult Accompanying Documents	NOTICE: This equipment shall be transported and stored under the following conditions: <u>Shipping</u> 1. Ambient temperature range of +5°C to +65°C. 2. Relative humidity range of 10% to 100%, non-condensing. <u>Storage</u> 1. Ambient temperature range of +10°C to +40°C. 2. Relative humidity range of 10% to 90%, non-condensing. Atmospheric pressure range of 500 hPa to 1060 hPa.
 Type BF Equipment	
 Class II Equipment	

WARNING: This product contains chemicals, including lead, known to the state of California to cause birth defects or other reproductive harm. **Wash hands after handling.**

Caution:
Federal law restricts this device to sale, distribution and use by or on the order of a physician.

INTRODUCTION

This manual describes the safety precautions, features, use and care of the Invivo 1.5T HD 4 Channel Wrist Array, for use with the GE Value MR System. Please review this manual thoroughly before using the device.

If you have any questions or comments regarding this manual, or need any assistance with the use of the product, please contact your GE Healthcare representative at:

1-800-GE CARES

COMPATIBILITY

The Invivo 1.5T HD 4 Channel Wrist Array, Invivo model number 800359, is compatible with the GE Value MR System. The minimum required system software level is 14.0.

EXPLANATION OF SYMBOLS



Medical Device Directive - These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the CE Mark of Conformity.



Attention, Consult Accompanying Documents.



Type BF Equipment



Class II Equipment



Do not cross or loop cables. Arcing and patient burns could result.



Waste in Electrical and Electronic Equipment



A hazard exists which could cause serious injury.

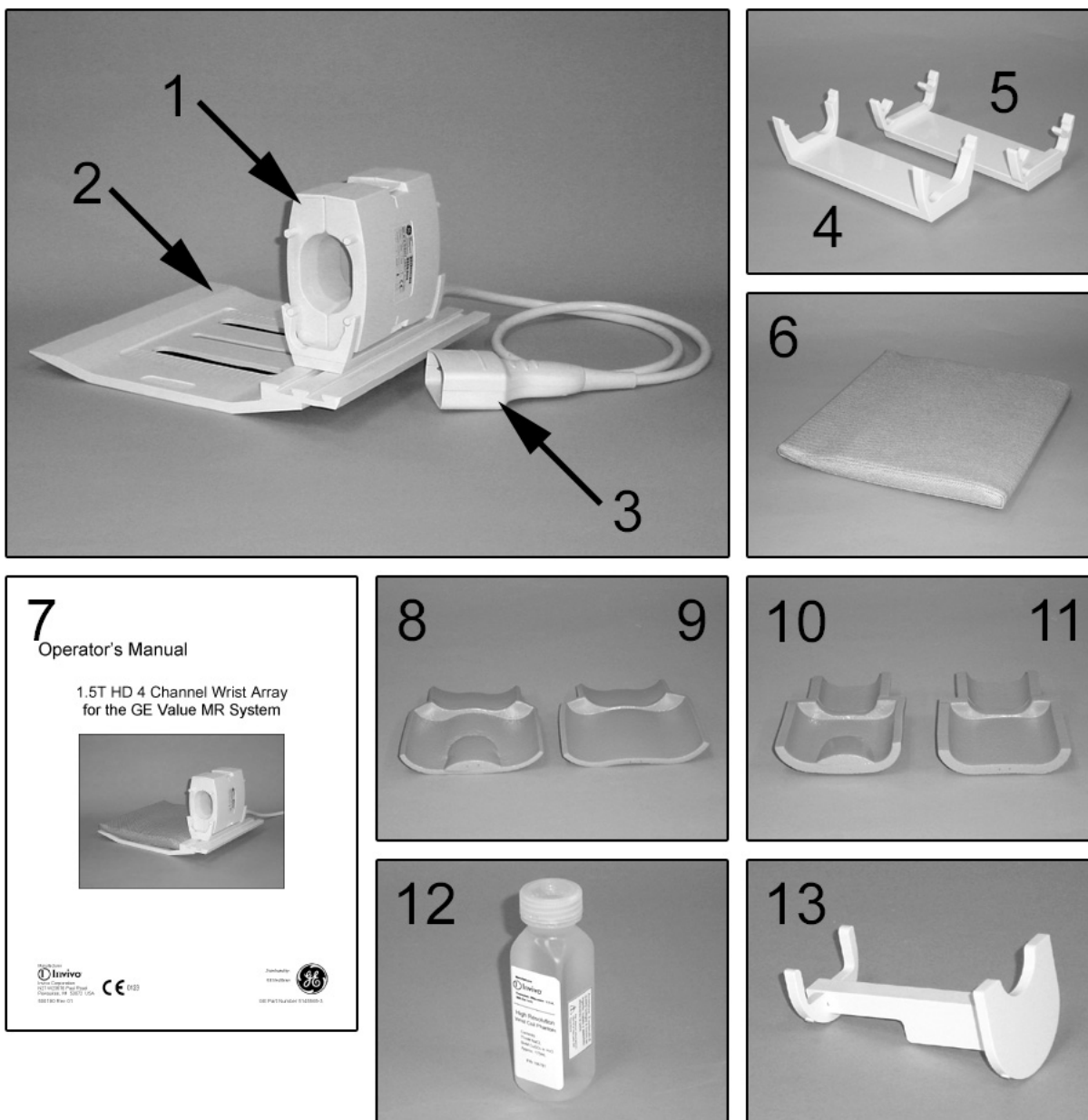


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CHAPTER 1 - 1.5T HD 4 CHANNEL WRIST ARRAY CONTENTS

The 1.5T HD 4 Channel Wrist Array consists of the following parts. Please inspect upon receipt to make sure all parts have arrived and are in good order. Use this guide to refer to part names throughout this manual.



- | | | |
|--------------------------|---------------------------|---------------------------|
| 1. 4 Channel Wrist Array | 6. Baseplate Pad | 11. Small Wrist Pad, Back |
| 2. Baseplate | 7. Operator's Manual | 12. Wrist Phantom |
| 3. Coil Cable Connector | 8. Large Wrist Pad, Palm | 13. Phantom Positioner |
| 4. Vertical Holder | 9. Large Wrist Pad, Back | |
| 5. Horizontal Holder | 10. Small Wrist Pad, Palm | |

CHAPTER 2 - SAFETY

2-1 Training

This manual contains detailed information regarding the set-up, positioning and use of the Invivo Corporation coil. The instructions must be read carefully and thoroughly before attempting to scan patients with this coil.

2-2 Quality Assurance

The procedure described in the Quality Assurance chapter of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

2-3 Indications

The coil is indicated for use, on the order of a physician, in conjunction with a MR scanner as an accessory to produce images of the wrist and hand.

2-4 Contraindications

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR system:

 Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants (cardiac pacemakers, for example). The magnetic and electromagnetic fields produced by the MR system and coil may interfere with the operations of these devices.

 Scanning patients with intracranial aneurysm clips is contraindicated.

2-5 Precautions

Precautions should be taken when scanning patients with the following conditions:

 Greater than normal potential for cardiac arrest

 An increased likelihood for developing seizures or claustrophobia

 Unconscious, heavily sedated, or confused patients

 Inability to maintain reliable communications

2-6 Cautions

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system Operator's Manual.

-   Do not cross or loop cables. Arcing and patient burns could result. Route cables out of the magnet so that they do not touch the patient.
-  Assure that the patient is not touching the bore. If necessary, place pads between the patient and the surface of the bore.
-  If the patient complains of warming, tingling, stinging, or similar sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.
-  Patients with implanted ferromagnetic metal should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.
-  Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.
-  There is a risk to scanning feverish or decompensated cardiac patients.
-  Facial makeup should be removed before scanning because it may contain metal flakes which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.
-  Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing a MR exam.
-  Visually inspect the cable insulator jackets, strain reliefs, and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.

2-7 Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

2-8 Technical Considerations

-  The coil and accessories require special conditions regarding electromagnetic compatibility. The coil must be installed and used in a shielded scan room provided with the MR magnet and system. The user must ensure that the scan room door is closed during system use. Failure to do so may cause reciprocal interference with portable and mobile RF communications equipment, affecting the performance of the MR coil and/or such equipment.
-  The coil should only be used with the accessories specified in the Operator's Manual.
-  The use of accessories other than those specified in the Operator's Manual may result in decreased ESD immunity of the coil or MR system, causing damage to the coil and/or system.
-  The equipment should not be used with other coils or equipment present in the MR scanner except as specified in the Operator's Manual.
-  Tampering with the cable pins and connector may damage the connector and affect coil or system performance. Please verify that connector and pins are not damaged before use.

CHAPTER 3 – QUALITY ASSURANCE

3-1 Tools Required

TOOLS REQUIRED

Description	GE Part #	Supplier Part #	Qty
Wrist Phantom	2293670-3	100885	1
Phantom Positioner	2293670-4	100868	1

3-2 Setup

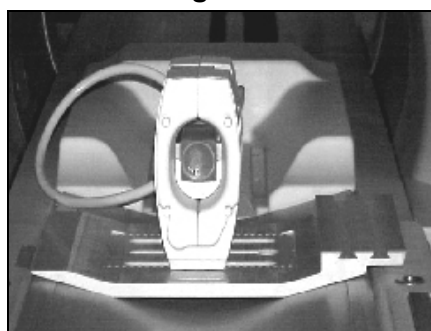
1. Remove all surface coils from the cradle. Position the High Resolution Wrist Coil in the vertical configuration on the center of the baseplate. Place the coil and baseplate on the patient cradle with the coil cable extending into the magnet [Figure 2].
2. Place the phantom on the phantom positioner and insert them into the coil as illustrated in Figures 1 and 2.
3. At the magnet, press the “Alignment Light” button to turn on the light. Move the cradle to align the coil to the alignment lights as shown in Figure 3. Press the “Landmark” button to landmark the alignment.
4. Move the coil to scan position by pushing the “Move to Scan” button, ensuring that the cable does not get snagged.

Figure 1



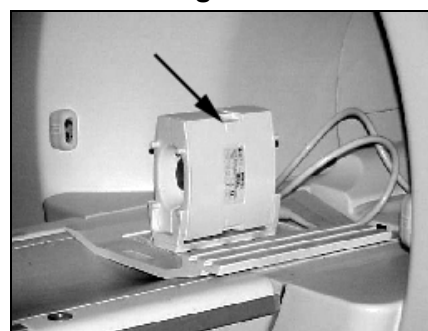
Phantom & Positioner

Figure 2



Coil Positioning

Figure 3



Coil Landmark

3-3 Procedure

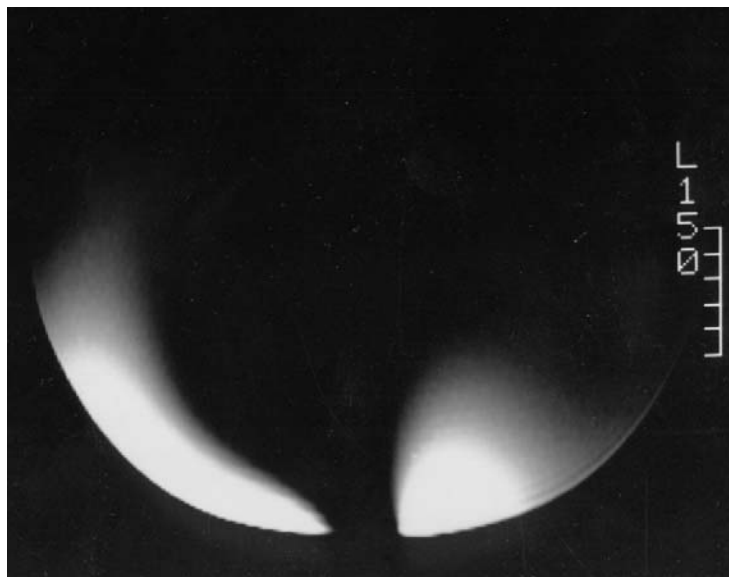
Be sure that you choose the correct coil, and activate the desired coils from the menu selection on the surface coil screen. The chosen coil must be HRWRIST. Activate the HRWRIST coil from the soft key on your operator's console.

Scan using Auto Prescan. If Auto Prescan does not complete successfully, do not use the coil clinically. Upon completion of Prescan, execute the scan. Window the scan as you normally would (window levels vary from system to system depending upon your hardware revision level). The scan should appear uniform.

If the image has signal voids or distortions as shown below, discontinue use of the 1.5T HD 4 Channel Wrist Array and contact your GE Healthcare Representative. If the image is excessively noisy, repeat the scan using the GE 5-inch general purpose coil (or alternate coil) supplied with your system. If the 5-inch coil image exhibits similar artifacts, the problem likely lies with the MRI system.

Using the wrist Phantom,
scan using the protocol
shown.

If the image appears as
shown, the Wrist coil is
malfunctioning.
Discontinue use of the coil
and contact your GE
Healthcare representative.



Mode: Multi
PSeq: ME

TR: 300
TE: 12.0 1/1

256x128/1.0 NEX

FOV: 12 cm
Thk: 10.0 mm
Imgs: 5/00:42
QUAD

CHAPTER 4 - USING THE 1.5T HD 4 CHANNEL WRIST ARRAY

4-1 Choosing the Best Coil Position

The Wrist Array Coil includes a horizontal holder for scans performed over the patient's head [Figure 1]. A vertical holder is also provided for scans performed at the patient's side [Figure 2], or over the patient's head [Overhead Vertical below]. Patient size, comfort and scan preference will dictate which position is used.

Overhead scans are performed head first, while at side scans are performed feet first. These configurations allow the coil cable to reach the coil port. The coil cable must point into the magnet.

Figure 1 – Overhead Position – Face Down

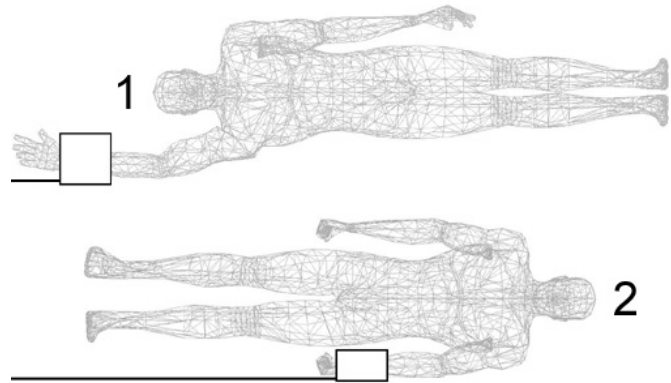
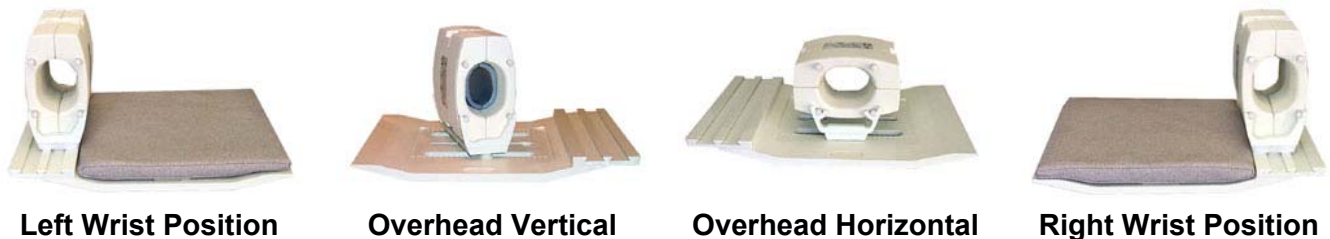


Figure 2 – At Side Position – Face Up



4-2 Positioning the Baseplate

The Wrist Array Coil includes a baseplate to support the coil holder and coil. The tray provides a flat stable surface upon which the coil can rest. Securing the coil greatly reduces motion artifact.

The baseplate is designed to rest directly on the patient cradle for stability. Position the baseplate near the magnet end of the patient cradle for overhead scans. For at side scans, position the baseplate near the middle of the patient cradle.



4-3 Locking the Coil Holder to the Baseplate



Figure 1 - At Side Position

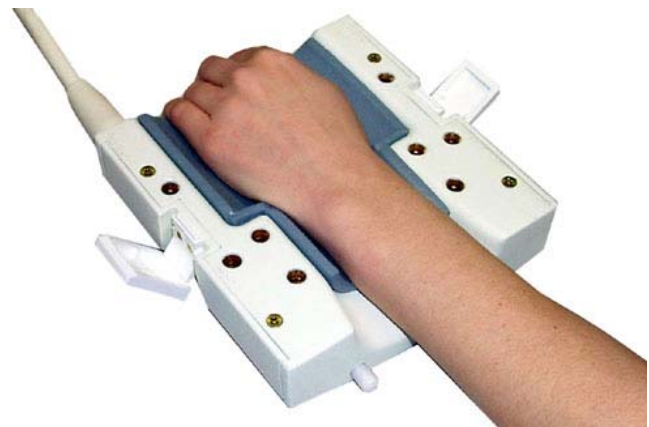


Figure 2 - Overhead Position

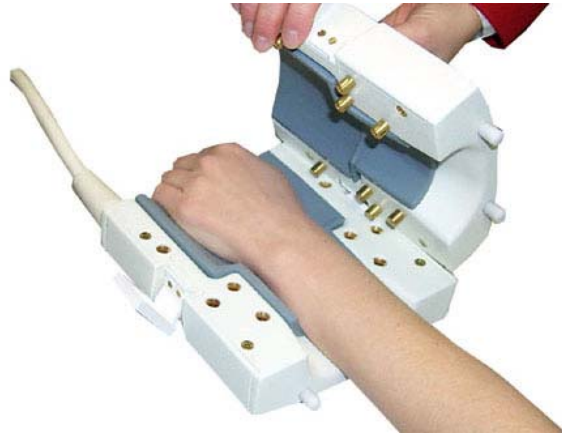
Once the patient position has been determined, select the proper coil holder for that position. For at side imaging, slide the holder into the slot as shown in Figure 1. When patient size permits, always use the medial slot to secure the holder. This moves the coil closer to the iso-center of the magnet for improved field homogeneity. For overhead imaging, slide the holder to the middle of the center slots as shown in Figure 2.

4-4 Patient Positioning

One thick and one thin pair of pads are provided to improve patient comfort and minimize image artifact caused by patient motion. Pads may not be necessary for large wrists. Select the appropriate pads and place the patient's wrist into the coil as illustrated on the right.

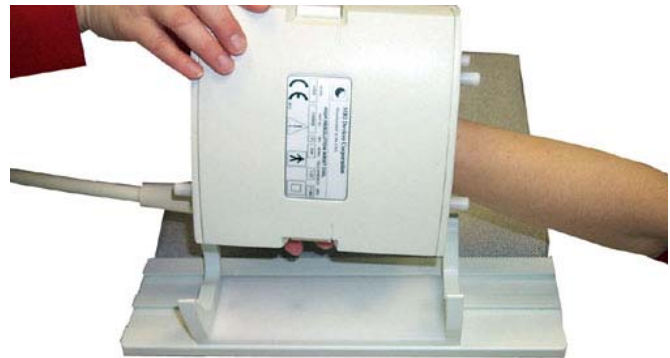


Using the connector pins as a guide, carefully close the coil halves. A snug fit is desired, so padding may be added or removed to achieve a good fit. Once the patient is comfortable, latch the halves together.



4-5 Locking the Coil to the Holder

Use the pins on each end of the wrist coil housing to snap the coil into place on the holder. Once in place, the coil and holder, together, may be adjusted in the slot for patient comfort.



4-6 Connecting the Cable

Connect the 1.5T HD 4 Channel Wrist Array to the system using the standard phased array connector.

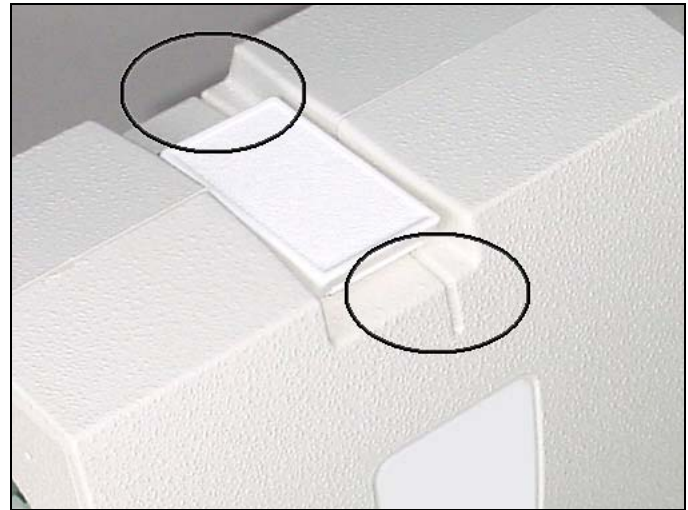


4-7 Landmark and Field of View

A landmark notch is provided in the latch recess of the coil housing. Landmark on the notch to center the FOV on the wrist.

The Wrist Array is 6 cm. high and 12 cm. wide and, therefore, has a maximum inherent field of view (FOV) of 12 cm. If desired, a smaller or larger FOV may be used.

Typically, axial wrist scans will be optimal with a 6 cm. to 8 cm. FOV. For imaging the hand, use up to a 10 cm. FOV. For high resolution imaging, 6 cm. FOV scans can easily be performed.



See Chapter 5 for suggested scan protocols.

4-8 Patient Hearing Protection

Provide ear plugs for the patient after all instructions have been given.



Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

Hearing protection must have a Noise Reduction Rating (NRR) of 28 dB or better (e.g., 30 dB, 32 dB, etc.).

CHAPTER 5 - SCANNING

5-1 Selecting the Coil

The coil description displayed to the user is HD 4Ch Wrist Array by Invivo. This coil mode uses all 4 elements of the coil.

5-2 Localizing with the Body Coil

The HRW coil is designed to allow the scanner body coil to be used while the HRW coil is in the scanner. This allows use of a large FOV (40 cm.) body coil localizer, which is helpful in determining the left or right offset required for imaging the wrist or hand joints.

You may scan using the body coil at any time.

5-3 Fat Saturation Techniques

Off-center FOV imaging is a more difficult technique since it is very dependent on the homogeneity of the magnetic field, and a definite fat peak signal. A small wrist on the edge of the imageable field may produce poor fat saturation.

Moving the coil toward the center of the bore may help when patient size allows. Even a centimeter or two is often helpful.

For axial imaging, use an axial localizer before prescribing slices, localize using the same FOV that you intend to use in your study, or localize explicitly. Use a of graphic prescription from a large FOV localizer sometimes results in software miscues, producing blank slices, shifted slices, or both.

5-4 Protocols

Refer to the GE protocols on your system.

CHAPTER 6 – INSTALLATION AND MAINTENANCE

6-1 Installation and Configuration

Coil installation and configuration require the services of your GE Service Representative. Please refer to the Service Manual for coil installation and configuration information.

6-2 Cleaning

The 1.5T HD 4 Channel Wrist Array, baseplate and pads must be cleaned using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

6-3 Storage

This equipment must be stored under the following conditions:

- Ambient temperature range of +10°C to +40°C
- Relative humidity range of 10% to 90%, non-condensing
- Atmospheric pressure range of 500 hPa to 1060 hPa

To store the coil and baseplate, a storage space of 35.6 cm./14.24 in. (width) x 43.8 cm./17.52 in. (depth) x 21.4 cm./8.6 in. (height) is required.

6-4 Replaceable Accessories

The following table lists replaceable accessories that may be purchased from Invivo Corporation:

Description	Invivo Part #
Baseplate Pad	100784
Small Wrist Pad, Palm	100864
Small Wrist Pad, Back	100865
Large Wrist Pad, Palm	100866
Large Wrist Pad, Back	100867

6-5 Contact Information

For additional information, contact:

GE Healthcare Americas
USA 800-558-5102
Canada 800-668-0732

GE Healthcare Asia
China 86-21-62192228
Taiwan 886-2-2505-7900
Singapore 65-291-8528
Australia 61-2-9975-5501
Japan 81-120-48-2630
Korea 82-31-740-6119
India 91-80-845-2923

GE Healthcare Europe
(33) 1-41-19-76-76

GE Healthcare Web Site
www.gehealthcare.com

Invivo Corporation
800-524-1476
262-524-1402

Invivo Web Site
www.invivocorp.com

CHAPTER 7 – ADDENDUM



注释：可替换的识别标签已随模提供，标签规定了模的使用方法，并附有多种语言版本。请依实际情况选择合适的标签直接贴于已有标签之上。

关于模的多语言模标签：106764

BEMÆRK: Skiftende identifikationsmærkater med håndteringsvejledning på flere sprog, leveres med dette fantomaggregat. Vælg den passende mærkat og placer den direkte ovenpå den eksisterende mærkat.

Flersproget mærkatark til fantom: 106764

N.B.: Bij deze fantoomassemblage zijn wisselende identificatielabels geleverd die in een verscheidenheid aan talen instructies geven voor hantering van het fantoom. Selecteer het juiste label en plaats dit direct bovenop het bestaande label.

Meertalig labelsheet fantoom: 106764

Note: Alternate identification labels have been supplied with this phantom assembly which provide phantom-handling instructions in a variety of languages. Please select the appropriate label and apply it directly on top of the existing label.

Multi-language phantom label sheet: 106764

HUOMAA: Tämän fantomikokoonpanon mukana on toimitettu erilaisia tunnusmerkkejä, joista löytyvät fantomin käsittelyohjeet eri kielillä. Valitse sopiva merkki ja kiinnitä se suoraan laitemerkin päälle.

Monikielinen fantomin merkkilappu: 106764

REMARQUE : d'autres étiquettes d'identification ont été fournies avec ce fantôme. Elles fournissent des instructions de manipulation du fantôme en plusieurs langues. Sélectionner l'étiquette appropriée et l'appliquer directement par-dessus l'étiquette existante.

Fiche de l'étiquette du fantôme multilingue : 106764

HINWEIS: Mit diesem Phantom wurden Wechseletiketten geliefert. Sie enthalten die Anleitung zur Handhabung des Phantoms in verschiedenen Sprachen. Bitte wählen Sie das passenden Etikett und überkleben Sie damit das vorhandene Etikett.

Mehrsprachiger Phantom-Etikettenbogen: 106764

ΣΗΜΕΙΩΣΗ: Αυτή η διάταξη phantom διατίθεται με εναλλακτικές ετικέτες επισημάνσης, οι οποίες παρέχουν οδηγίες χειρισμού του phantom σε διάφορες γλώσσες. Επιλέξτε την κατάλληλη ετικέτα και τοποθετήστε την απευθείας επάνω από την υπάρχουσα ετικέτα.

Φύλλο ετικετών phantom πολλαπλών γλωσσών: 106764

NOTA: Sono incluse etichette che forniscono le istruzioni di utilizzo in diverse lingue. Selezionare l'etichetta adeguata e applicarla direttamente sopra all'etichetta esistente.

Etichette per fantoccio in lingue diverse: 106764

注意： 代替識別ラベルが、様々な言語で記述されたファントム使用説明書付きのファントムアセンブリと共に提供されます。適切なラベルを選択し、既存のラベルの上に直接貼ってください。

多国言語対応ファントムラベルシート：106764

참고: 이 팬텀 어셈블리와 함께 제공된 대체 ID 레이블에서는 팬텀 취급 지침을 다양한 언어로 전달합니다. 알맞은 레이블을 선택하여 기존 레이블 바로 위에 부착하십시오.

다국어 팬텀 레이블 시트: 106764

MERK: Det medfølger alternative identifikasjonsmerker med denne fantomenheten, som inneholder instruksjoner for fantomhåndtering på flere språk. Velg det passende merket, og påfør det direkte oppå det eksisterende merket.

Flerspråklig fantommerkeark: 106764

UWAGA: Z niniejszym fantomowym zestawem zastępczym dostarczane są alternatywne etykiety identyfikacyjne, zawierające instrukcje w różnych językach. Proszę wybrać stosowną etykietę i umieścić ją na wierzchu istniejącej na zestawie.

Arkusz z etykietami zastępczymi w wielu językach: 106764

NOTA: Juntamente com este fantoma são fornecidos rótulos de identificação alternativos, contendo instruções para a manipulação do fantoma, em vários idiomas. É favor seleccionar o rótulo apropriado e aplicá-lo directamente por cima do rótulo existente.

Folha com os rótulos de fantoma em vários idiomas: 106764

OBSERVAȚIE: etichetele de identificare alternative au fost livrate împreună cu acest ansamblu al fantomei și conțin instrucțiuni de manevrare în diferite limbi. Selectați eticheta corespunzătoare și aplicați-o direct peste cea existentă.

Fisă multilingvistică de etichetare a fantomei: 106764

ПРИМЕЧАНИЕ. С этим фантомным блоком поставляются запасные идентификационные наклейки, которые содержат инструкции по работе с фантомом на разных языках. Выберите соответствующую наклейку и наклейте ее поверх инструкций на блоке.

Таблица многоязычных наклеек для фантома: 106764

Nota: Con este fantoma se suministran etiquetas alternativas de identificación con instrucciones para la manipulación del fantoma en varios idiomas. Por favor seleccione la etiqueta que corresponda y aplíquela directamente sobre la etiqueta existente.

Hoja multilingüe de etiquetas del fantoma: 106764

OBS: De alternativa identifikationsetiketter som medföljer denna fantomenhet tillhandahåller fantomhanteringsinstruktioner på flera olika språk. Välj lämplig etikett och placera den direkt ovanpå befintlig etikett.

Ark med etiketter på flera språk: 106764

NOT: Bu fantom tertibatla birlikte çeşitli dillerde fantom kullanma talimatı veren alternatif tanımlama etiketleri sağlanmıştır. Lütfen uygun etiketi seçin ve doğrudan mevcut etiketin üzerine uygulayın.

Çeşitli dillerde fantom etiketi föyü: 106764

REVISION HISTORY

Revision	Date	Change Description	DCN
01	02-05-06	Initial Issue	N/A
02	05-31-06	Addition of Swedish Language Manual Added Swedish Note in Addendum Revised Section 3-3 Invivo Address was Pewaukee Added Revision History	2090
03	11-09-06	Corrected EU Representative, Page 2	2297
04	06-18-07	Added Korean warning note in Addendum	2992
05	02-19-08	Add 9 languages to Operator's Manual package. Add language identifier in top right corner of each translated cover page.	3648

